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AgriVision

Precision Agriculture & Autonomous AI Advisory Platform

Empowering farmers with real-time IoT ground sensors, satellite earth observation, and AI-driven agronomic intelligence.

NATIVE IOS MOBILE APP

WEB GIS AGRONOMIST PORTAL

FINAL YEAR PROJECT (FYP) PRESENTATION

SECTION 1: PROJECT INTRODUCTION

Executive Overview & Platform Vision

Bridging Modern Technology and Everyday Farming

Smallholder Farmers

Get plain-language actionable advice on irrigation, fertilization, and crop health directly on mobile devices without complex jargon.

Agronomists & Experts

Supervise hundreds of fields remotely using high-resolution spectral satellite maps, live telemetry, and AI safety triage queues.

Tri-Source Engine

Seamlessly fuses ground IoT soil probes, Sentinel-2 earth observation, and smartphone diagnostic photos into one unified assistant.

SECTION 2: BACKGROUND OF PROBLEM

The Smallholder Farming Crisis

High Input Costs and Inefficient Resource Usage

Water Wastage

Outdated flood irrigation practices cause up to 40% water loss, severe groundwater depletion, and nutrient runoff across small farms.

Over-Fertilization

Applying expensive fertilizers without soil testing burns crop roots, degrades long-term soil health, and inflates operational costs.

Unpredictable Weather

Sudden heatwaves and localized micro-climate shifts leave farmers vulnerable to sudden yield reduction and unexpected losses.

SECTION 2: BACKGROUND OF PROBLEM

Lack of Timely Expert Guidance

Why Farmers Don't Get Expert Help When They Need It

1,500 : 1 Expert Ratio

Severe shortage of agricultural officers forces over 1,500 farming families to depend on a single extension worker for seasonal advice.

Delayed Diagnosis

Pest infestations and leaf blight are identified days or weeks too late, resulting in irreversible crop destruction before treatment.

Generic Blanket Advice

Farmers receive general regional radio broadcasts rather than field-specific recommendations tailored to their unique soil and crop state.

SECTION 2: BACKGROUND OF PROBLEM

Disconnected Data and High Complexity

Scientific Tools Are Not Built for Everyday Farmers

Data in Silos

Satellite imagery, local weather forecasts, and laboratory soil reports exist in isolated, technical portals that do not communicate.

Confusing Jargon

Complex scientific values like NDVI index fractions or Volumetric Water Content (VWC %) are incomprehensible to non-technical growers.

Unsafe Generic AI

Standard AI chatbots lack agricultural context, frequently hallucinating dangerous chemical chemical dosages that destroy crops.

SECTION 3: SOLUTION OF THE PROBLEM

Tri-Source Intelligent Farming

Connecting Macro Satellite Data with Micro Ground Telemetry

1. Satellite Observation

Sentinel-2 earth observation tracks vegetative growth (NDVI) and moisture stress across entire field parcels from orbit.

2. Ground IoT Probes

Physical in-situ microcontrollers measure real-time soil moisture and root zone temperature every 5 seconds.

3. Smartphone Visual Diagnostics

Farmers capture high-resolution leaf photos for immediate AI crop disease classification and symptom tracking.

Soil Sensor


Soil NPK sensor
Power supply: □9-24V DC
Output: □0-5V □0-10V □ 4-20mA
□485
Range: □N□P□ K

SECTION 3: SOLUTION OF THE PROBLEM

AI Agronomist in Every Farmer's Pocket

Translating Complex Sensor Data into Simple Action

Plain Actionable Messages

Translates complex sensor matrices into plain tasks: *"Soil is dry in North Field — irrigate 2 inches this evening."*

Multimodal Gemini Diagnostics

Leverages Google Cloud Vertex AI to identify visual leaf diseases (rust, blight, mildew) directly from mobile uploads.

Long-Term Season Memory

Maintains a rolling narrative history of past weather, irrigation events, and crop treatments across the full growing cycle.

SECTION 3: SOLUTION OF THE PROBLEM

Clinical Safety & Expert Verification

Ensuring AI Never Gives Harmful Chemical Advice

1. Automated Safety Gate

Any AI reasoning output containing fungicides, pesticides, or chemical dosage recommendations is automatically intercepted by rule-based classifiers.

2. Agronomist Review Queue

Intercepted recommendations enter the web GIS triage queue where certified agricultural specialists inspect field data and approve or adjust dosages.

3. Verified Mobile Push

Farmers only receive high-risk chemical recommendations after human expert validation, ensuring complete clinical peace of mind.

SECTION 4: DB DIAGRAM

Data Persistence Strategy

Relational, Geospatial, and Time-Series Data Management

PostgreSQL 16 Relational Core

Securely manages user identities, field ownership, farm records, expert review queues, and clinical audit trails.

PostGIS Spatial Engine

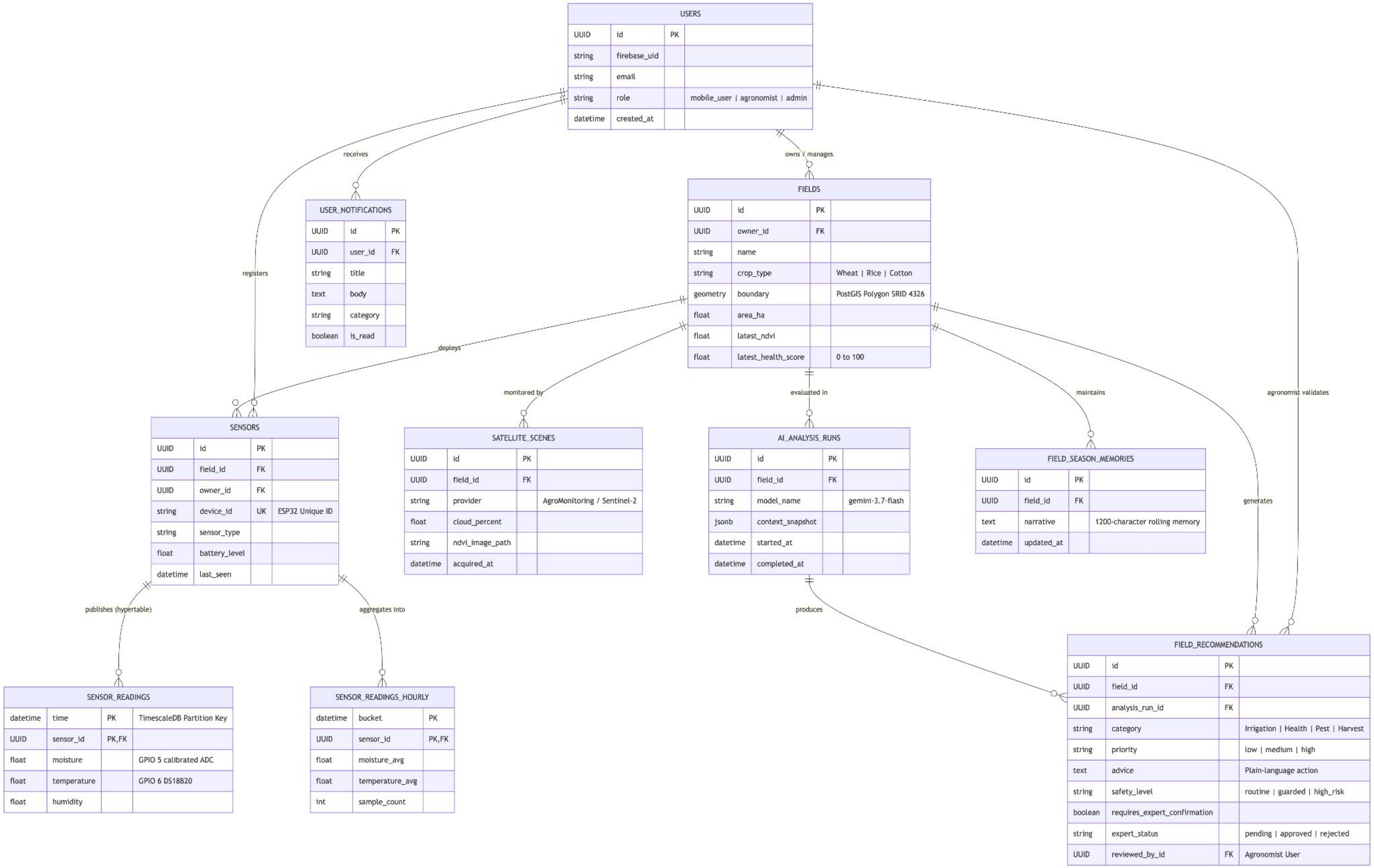
Stores geometric land boundary polygons (SRID 4326), calculates exact farm acreage, and handles spatial indexing queries.

TimescaleDB Hypertables

Optimized time-series engine for high-frequency IoT sensor telemetry with automated hourly rollups and chunk management.

Platform DB Diagram

Relational Mapping Across Users, Fields, Sensors, and AI Advisory



SECTION 5: ARCHITECTURE DIAGRAM

End-to-End System Workflow

How Information Flows from the Soil to the Screen

1 Sense & Capture

ESP32 Soil Nodes
Moisture & temperature via MQTT

Sentinel-2 Orbit
10m spectral NDVI tiles

Mobile Camera
Leaf symptom photo upload

2 Ingest & Sync

Mosquitto Broker
Async MQTT telemetry queue

FastAPI Gateway
Fusion & PostGIS indexing

TimescaleDB
Hypertables time-series storage

3 AI & Clinical Safety

Vertex AI Gemini
Agronomic reasoning engine

Safety Interceptor
Chemical advice gate filter

Agronomist Queue
Expert review & sign-off

4 Farmer Action

iOS Mobile App
Plain actionable alert:
"Irrigate field 2 today"

SECTION 5: ARCHITECTURE DIAGRAM

Multi-Tier System Architecture

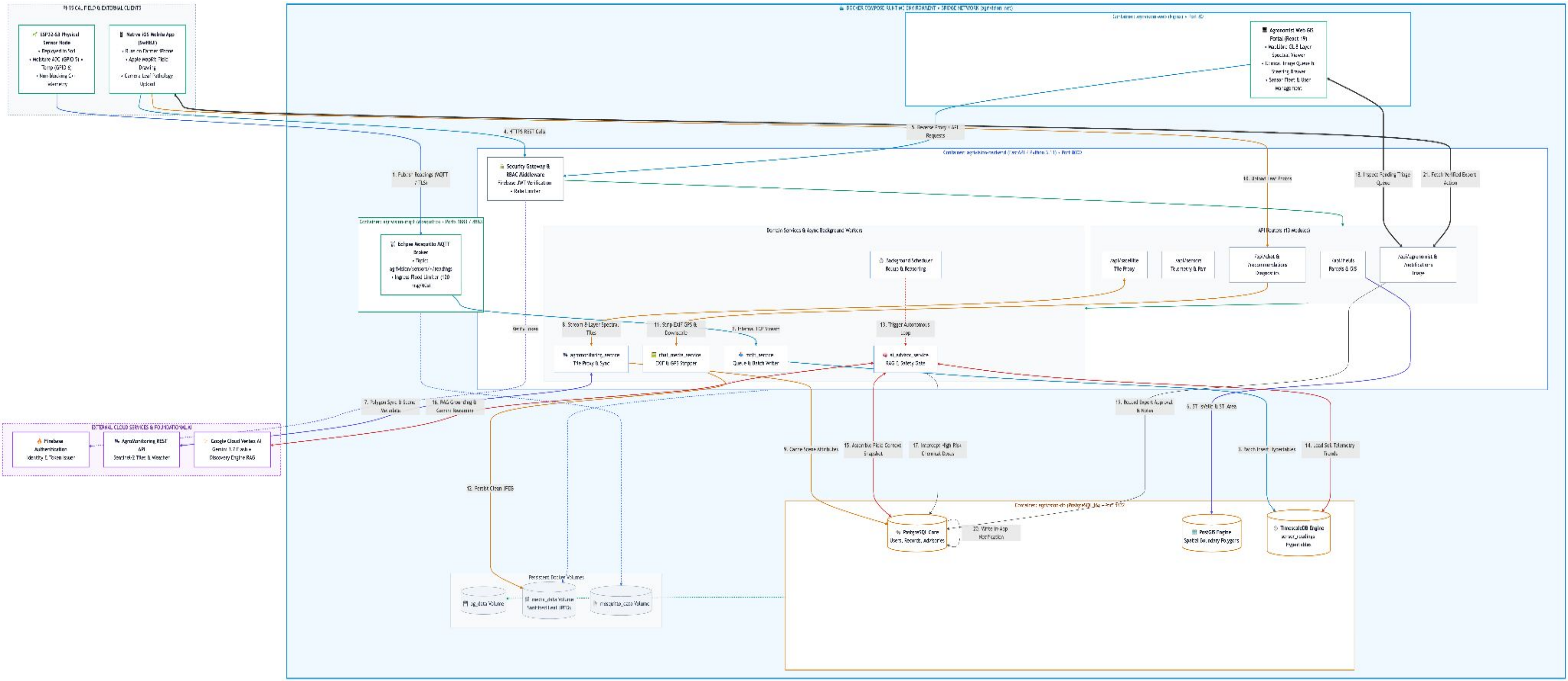
Clean, Scalable, and Decoupled Architecture

1. PRESENTATION	**Native iOS App** SwiftUI & Apple MapKit	**Agronomist Web Portal** React 19 & MapLibre GIS
2. EDGE IOT	**ESP32-S3 Firmware** GPIO Moisture & DS18B20 Temp	**Mosquitto Broker** Port 8883 MQTT / TLS
3. API GATEWAY	**FastAPI Async Router** Python 3.11 REST Endpoints	**Auth & Rate Limiting** Firebase JWT Verification
4. SERVICES & DB	**PostgreSQL + PostGIS** Spatial parcel boundaries	**TimescaleDB** Hypertable sensor metrics
5. CLOUD SERVICES	**Google Vertex AI** Gemini 3.7 Flash RAG Engine	**AgroMonitoring API** Sentinel-2 Spectral Tiles

SECTION 5: ARCHITECTURE DIAGRAM

Multi-Tier System Architecture

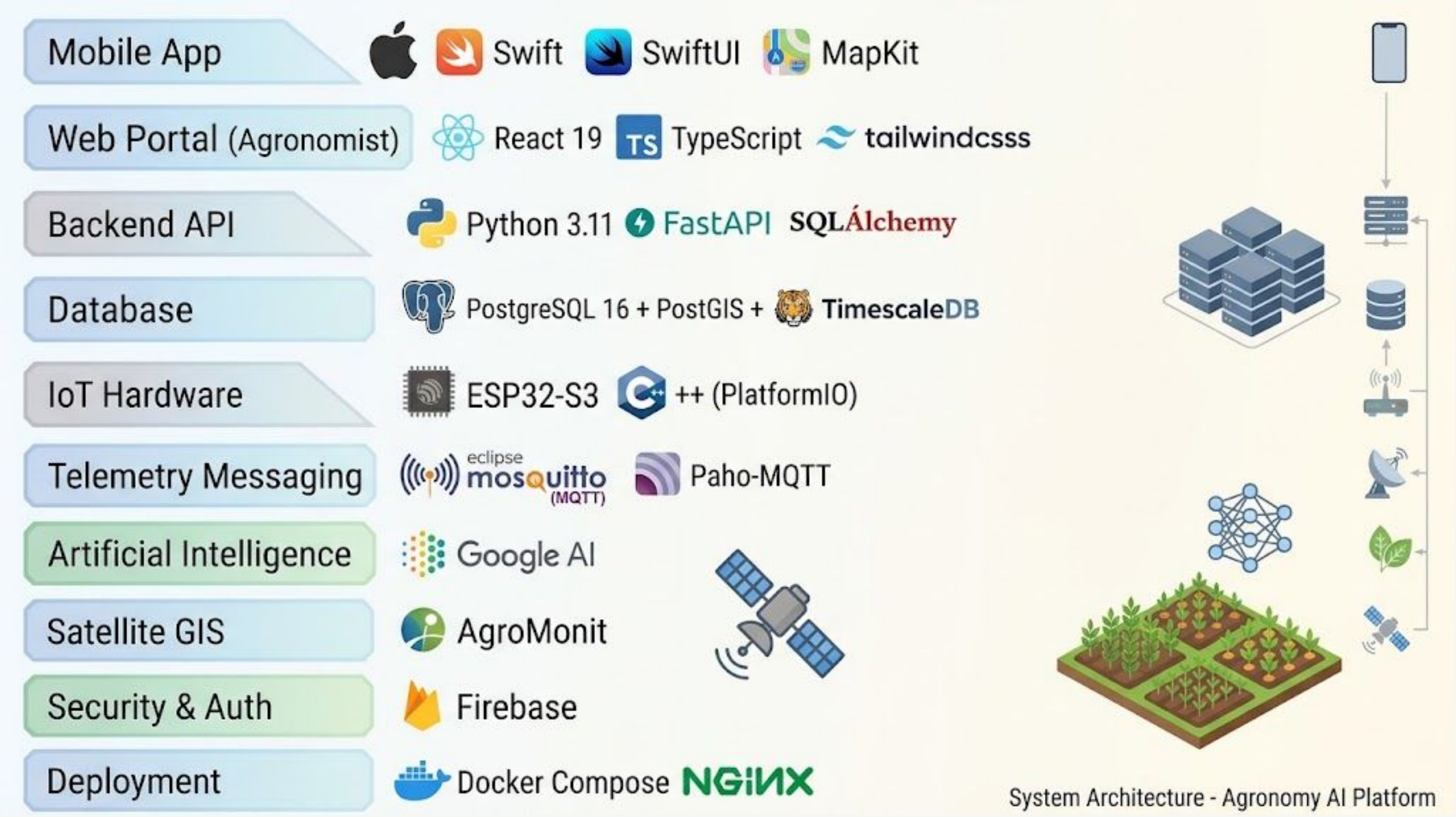
Clean, Scalable, and Decoupled Architecture



SECTION 6: TECHNOLOGY STACK

Frontend, Backend & Database Stack

Modern, Reliable Technologies Powering AgriVision



Interactive Field Boundary Mapping

Easy Land Digitization for Smallholder Farmers

Map Drawing Canvas

Farmers simply tap boundary points directly on satellite imagery to define precise field perimeters.

Automated Acreage Calculation

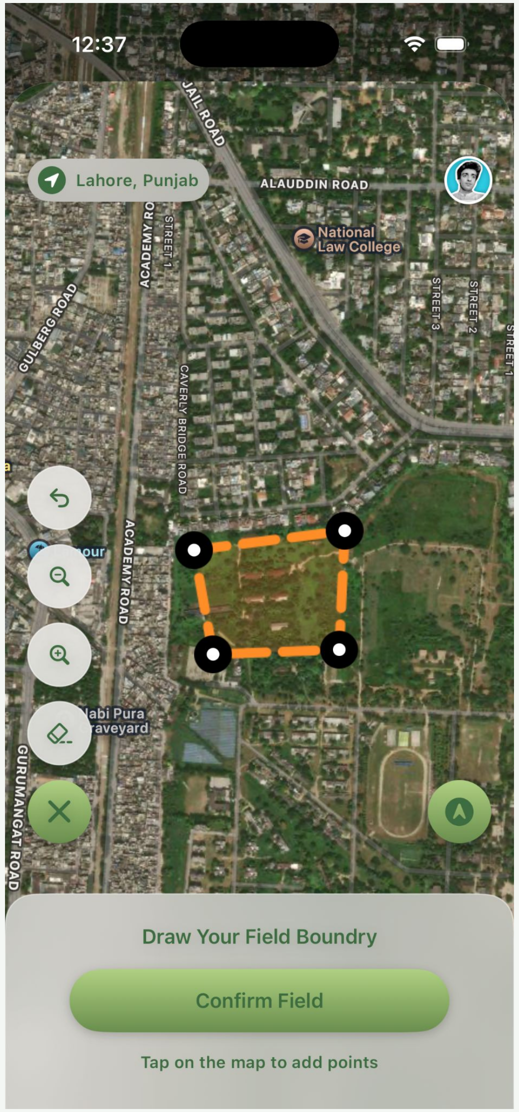
PostGIS back-end instantly calculates surface area in hectares and acres upon polygon closure.

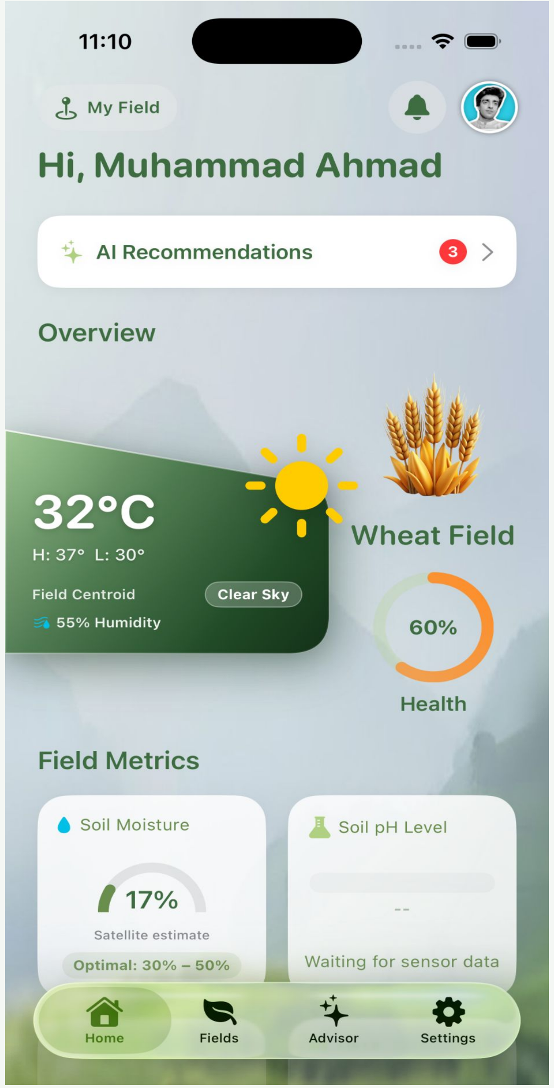
Crop Profile Configuration

Quick setup wizard selects crop type (Wheat, Rice, Sugarcane) and planting date for growth tracking.

SECTION 7: USER INTERFACES

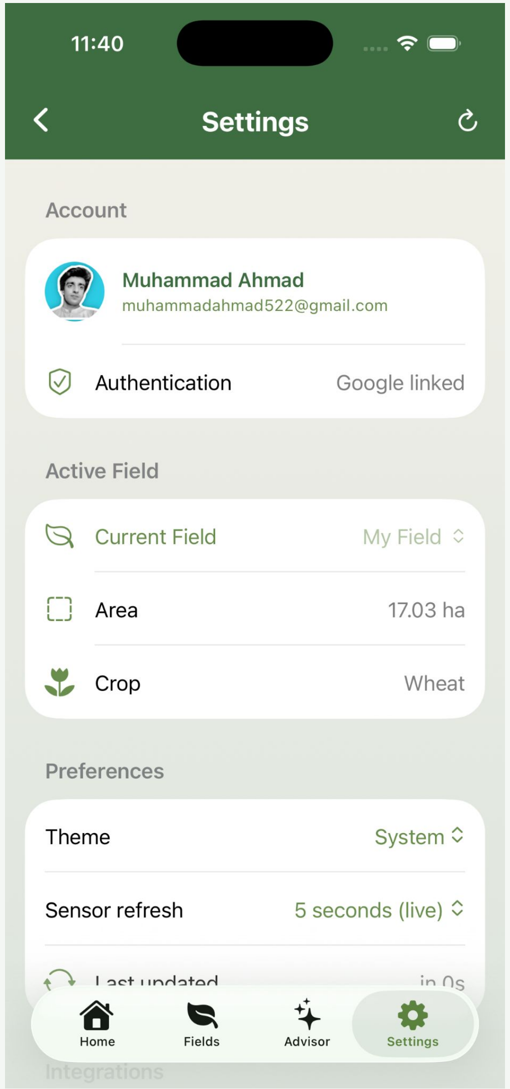
App Screens

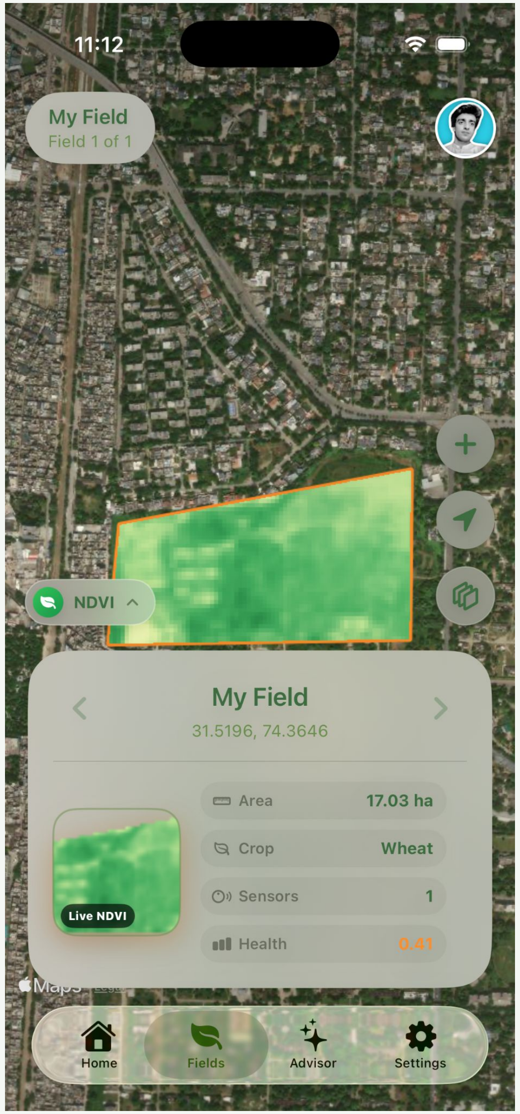


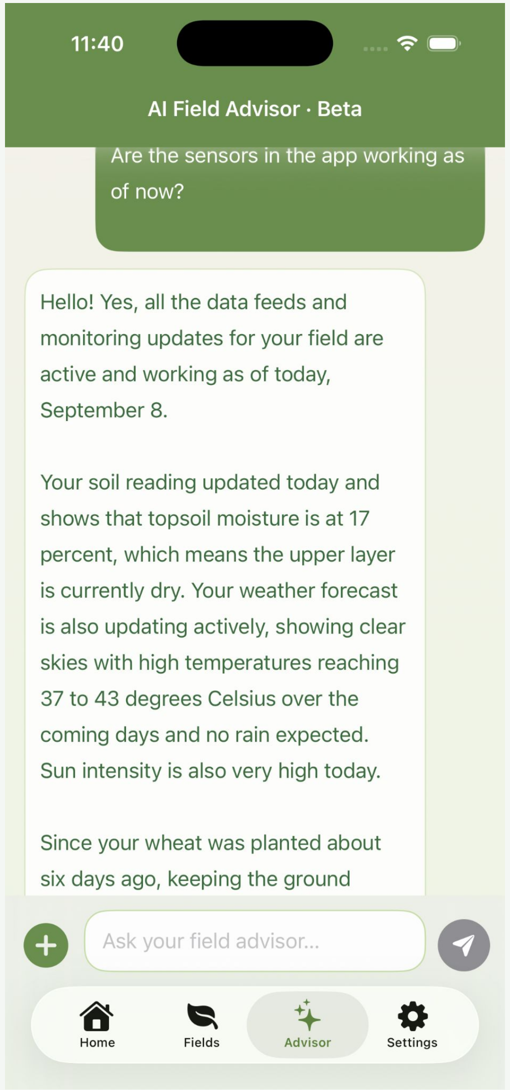


SECTION 7: USER INTERFACES

Ios App Field



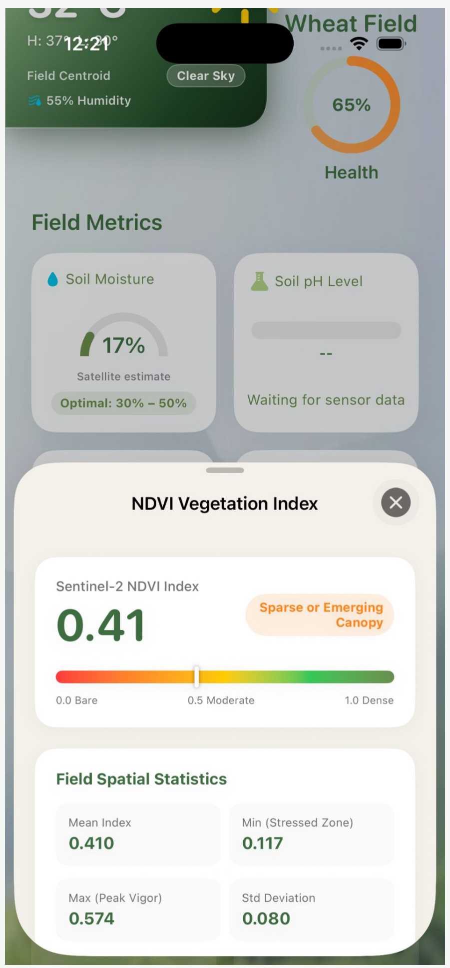


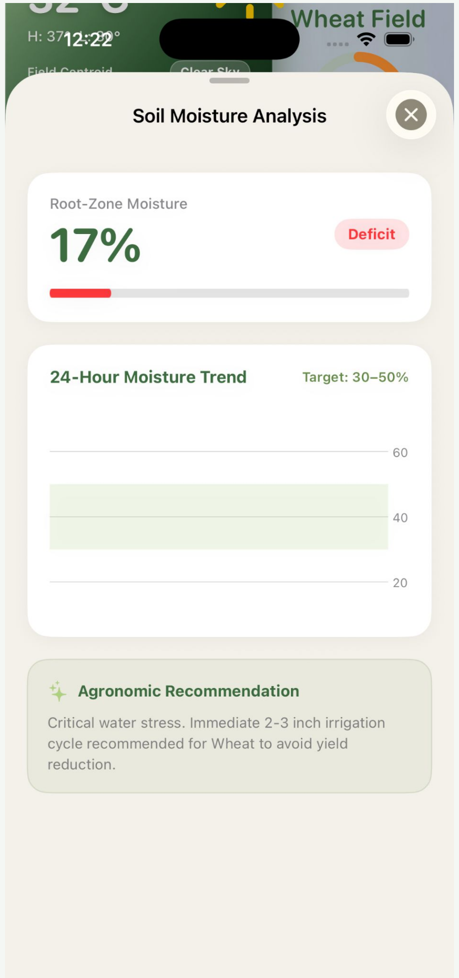


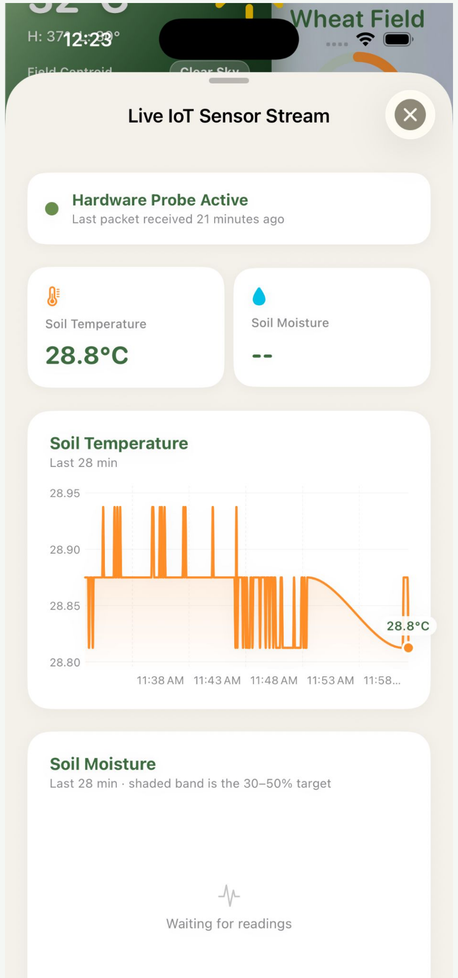
SECTION 7: USER INTERFACES

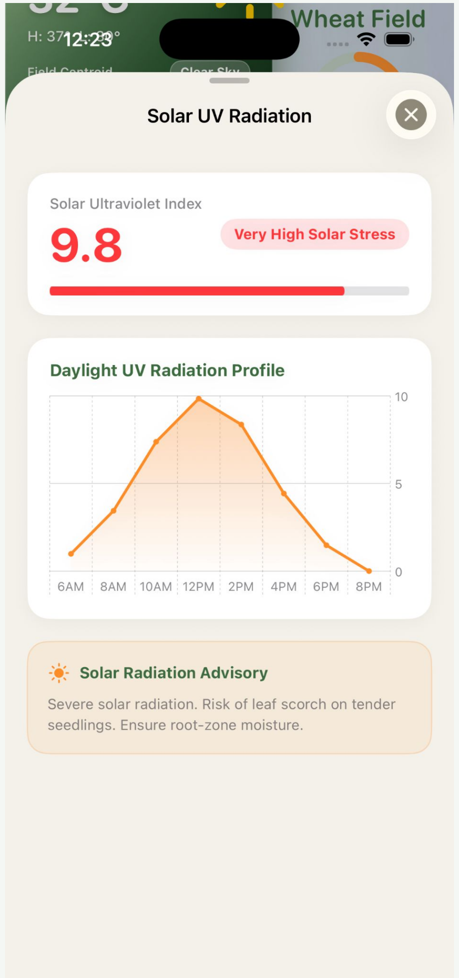
Field Health Dashboard & Live Metrics

All Essential Field Data in Clean Glanceable Cards



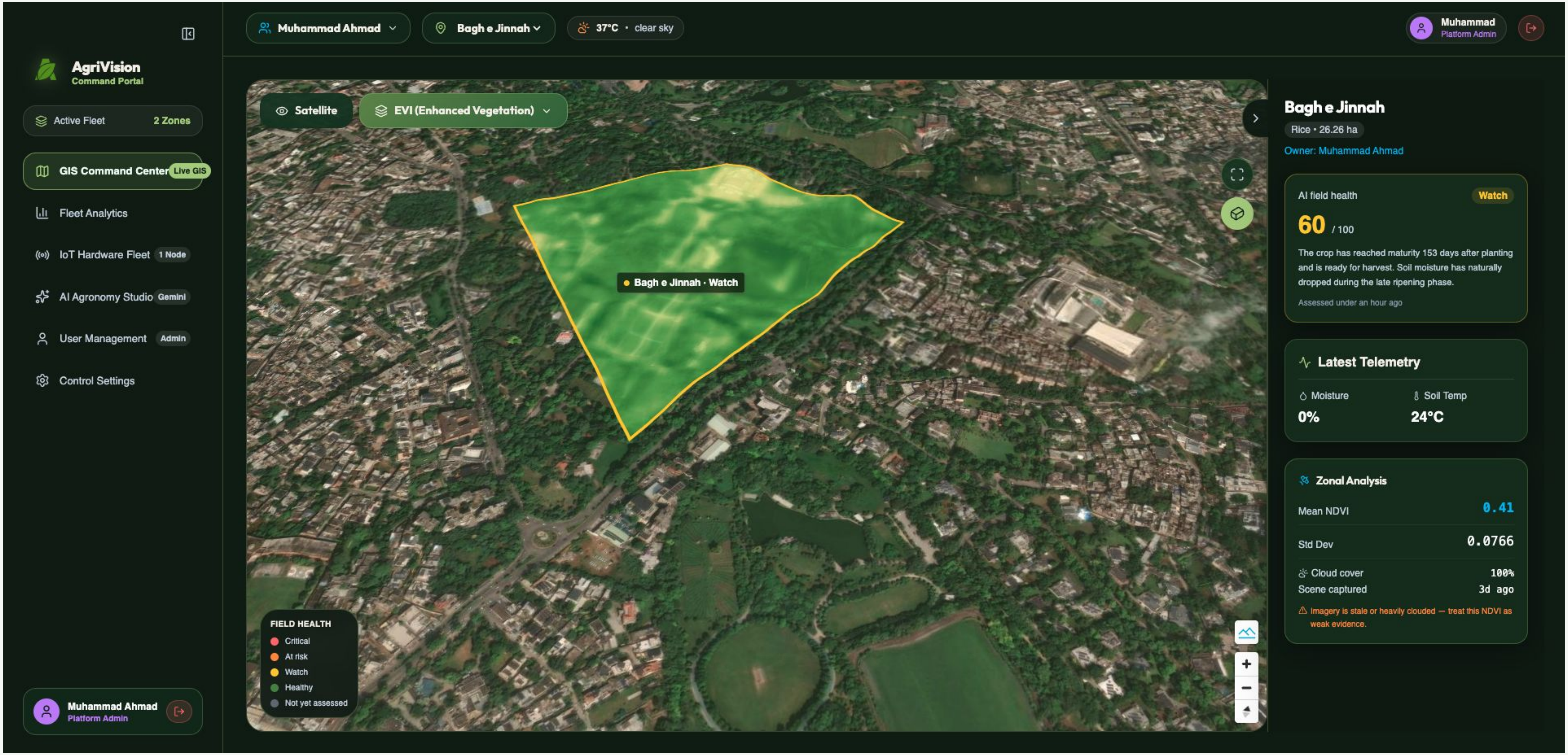




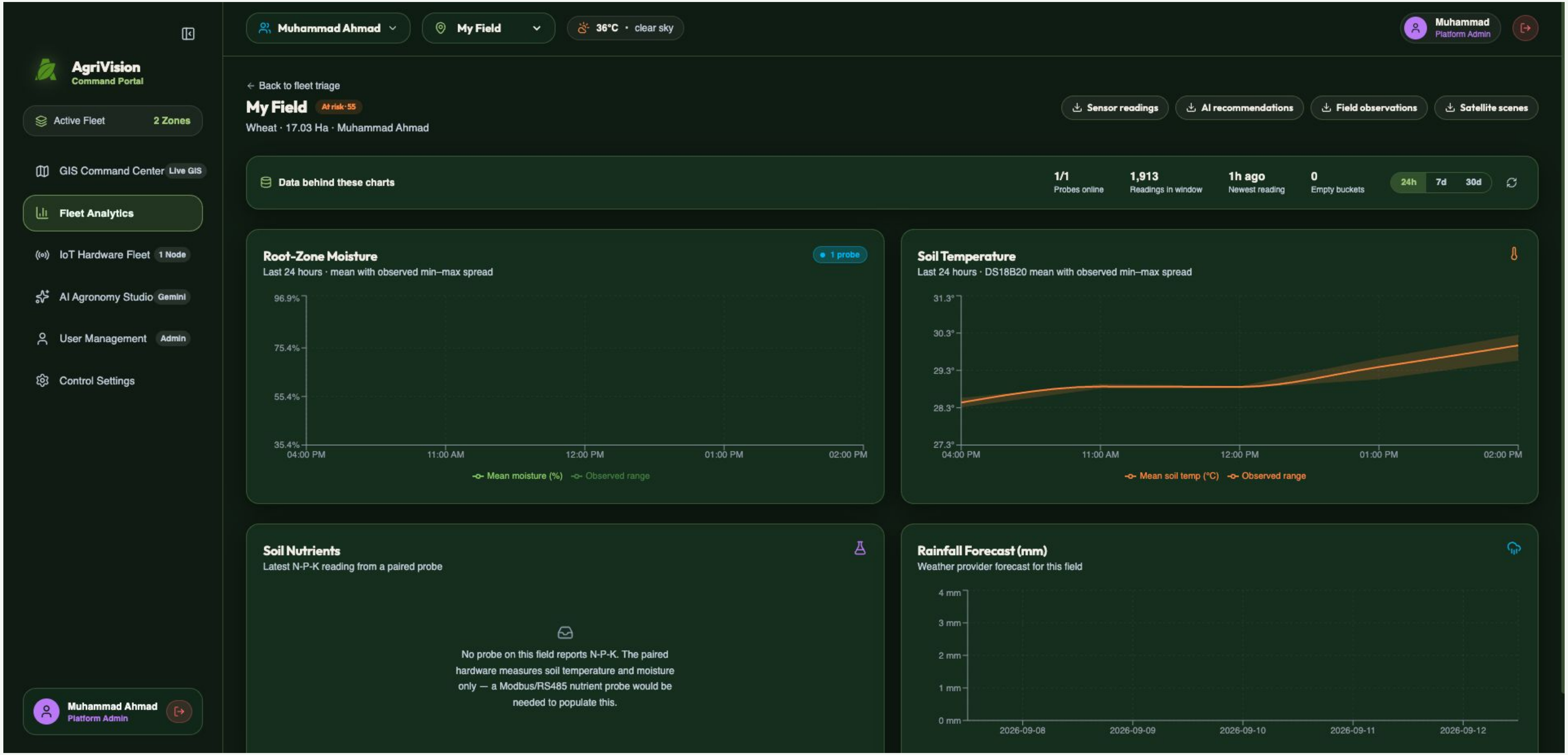


SECTION 7: USER INTERFACES

App Screens



App Screens


AgriVision
Command Portal
Active Fleet 2 Zones
GIS Command Center Live GIS
Fleet Analytics
IoT Hardware Fleet 1 Node
AI Agronomy Studio Gemini
User Management Admin
Control Settings
Muhammad Ahmad
Platform Admin
Muhammad Ahmad
My Field
36°C · clear sky
Muhammad
Platform Admin
← Back to fleet triage
My Field At risk · 55
Wheat · 17.03 Ha · Muhammad Ahmad
Sensor readings
AI recommendations
Field observations
Satellite scenes
Data behind these charts
1/1 Probes online
1,913 Readings in window
1h ago Newest reading
0 Empty buckets
24h 7d 30d
Root-Zone Moisture
1 probe
Last 24 hours · mean with observed min–max spread
96.9%
75.4%
55.4%
35.4%
04:00 PM 11:00 AM 12:00 PM 01:00 PM 02:00 PM
Mean moisture (%)
Observed range
Soil Temperature
Last 24 hours · DS18B20 mean with observed min–max spread
31.3°
30.3°
29.3°
28.3°
27.3°
04:00 PM 11:00 AM 12:00 PM 01:00 PM 02:00 PM
Mean soil temp (°C)
Observed range
Soil Nutrients
Latest N-P-K reading from a paired probe
No probe on this field reports N-P-K. The paired hardware measures soil temperature and moisture only — a Modbus/RS485 nutrient probe would be needed to populate this.
Rainfall Forecast (mm)
Weather provider forecast for this field
4 mm
3 mm
2 mm
1 mm
0 mm
2026-09-08 2026-09-09 2026-09-10 2026-09-11 2026-09-12

SECTION 8: CONCLUSION

Tangible Agricultural Impact

Delivering Real Value to Smallholder Communities

Water Conserved

Eliminates wasteful flood over-irrigation

Cost Savings

Reduced fertilizer & chemical wastage

Earlier Warning

Detects moisture & crop stress early

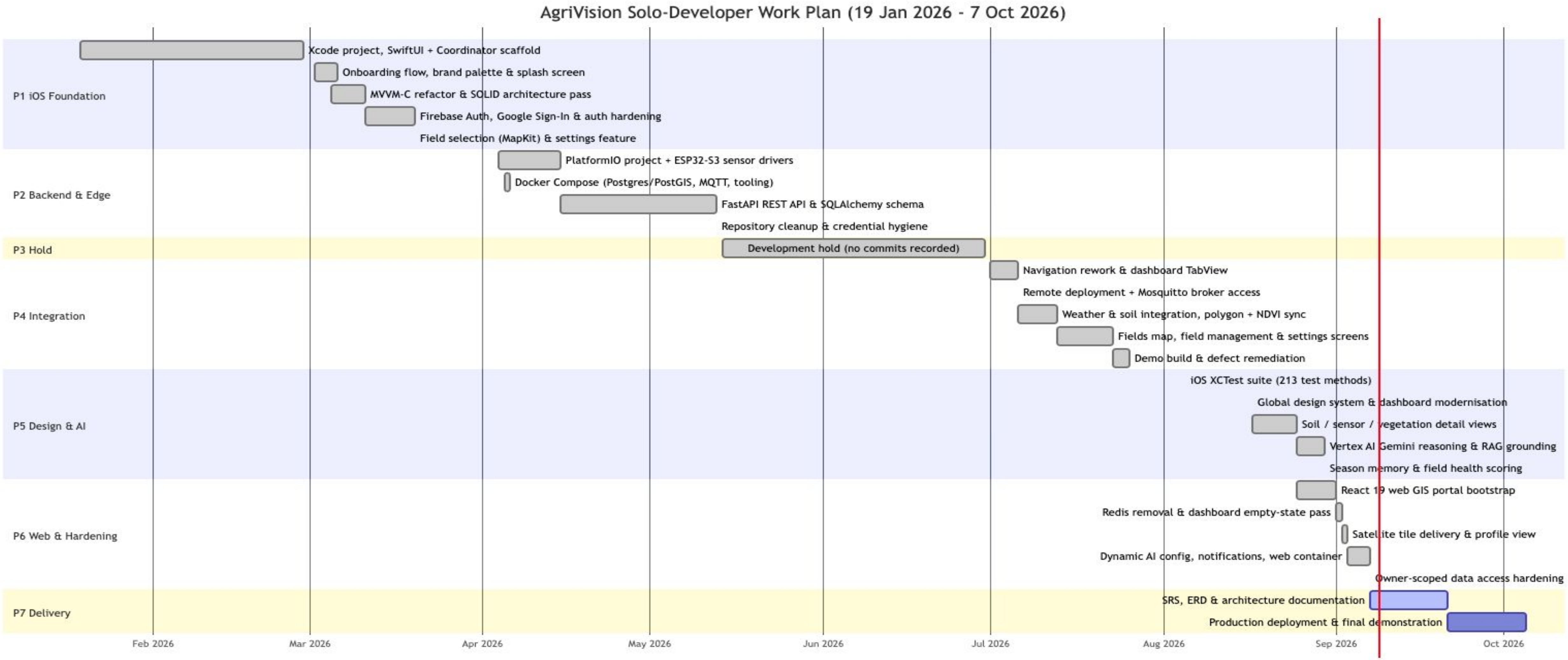
Safety Verified

Human-in-the-loop chemical protection

SECTION 8: CONCLUSION

Project Schedule and Gantt Chart

Delivering Real Value to Smallholder Communities



SECTION 8: CONCLUSION

Summary & Future Roadmap

Scalable Foundation for Autonomous Precision Agriculture

Future Expansion Roadmap

- → Direct push notifications for critical frost and heatwave emergency warnings.
- → Automated Over-The-Air (OTA) firmware update manager for deployed ESP32 field nodes.
- → Expanding multimodal leaf disease vision models to additional regional fruit & vegetable crops.
- → Closed-Loop Automated Valve & Irrigation Control and fertilization
  - Automatically triggers water delivery when soil moisture hits critical thresholds, eliminating manual pump operation.

  Automated drones will used for spraying and fertilization
- → In app purchasing of pesticides and chemicals

Key Accomplishments

- ✓ Successfully unified ground IoT, satellite imagery, and generative AI into working iOS and Web applications.
- ✓ Implemented a zero-hallucination clinical safety filter with human-in-the-loop agronomist triage.
- ✓ Built high-performance spatial PostGIS & TimescaleDB time-series storage pipeline.